Assignment - 5

⇒ Aim

Develop a web application for Employee Registration using JSP, Servlet & JDBC.

⇒ Objectives

- To understand server-side scripting
- To understand the concept of Servlet & JSP.
- To learn the working of servlet & JSP
- To lear the db connectivity using MySQL

⇒ Theory

1) Servlet :-

A servlet is a Java class used to extend the capabilities of servers that host applications accessed by means of a request-response programming model. It acts as the "Controller" handling logic & routing.

2) JSP (Java Server Pages)

JSP is a server-side technology that makes creating dynamic web pages easier by allowing Java code to be embedded within HTML.

It acts as the "View" in the MVC pattern.

3) JDBC [Java DB Connectivity]
JDBC is a Java API that manages the connection between the Java application & a database (in this case, MySQL). It allows for executing SQL queries & retrieving results within Java code.

⇒ Implementation

Step 1 - Database Setup

Step 2 - The view

Step 3 - The Data Access Object

Step 4 - The controller

⇒ Conclusion
Throug this assignment, implemented a web application that demonstrates server-side scripting using Servlets, dynamic content rendering with JSP, & persistent data storage via JDBC & MYSQL.

⇒ FAQs

1) What is the difference between Servlet & JSP?

→ • Servlet -
A Java-based program where HTML code is embedded inside Java code.
It is better for processing logic.

• JSP -
A text-based document where Java code is embedded inside HTML.
It is better for presentation.

2) List out advantages of using JSP.

→ • Better performance due to pre-compilation.
• Separation of business logic from presentation.
• Implicit objects are available, reducing the amount of code written.

3) What is the difference between Get & Post methods?

→ • GET -
Appends data to the URL, making it less secure & limited in size.

• POST -
Sends data in the HTTP request body, making it more secure & capable of handling large amounts of data.

4) Compare GenericServlet & HTTpServlet.

→ • GenericServlet –
Protocol – independent, must implement the service() method.

• HttpServlet –
Protocol – dependent (HTTP), provides specific methods like doGet() & doPost()

5) What are the major components of JDBC?

→ • DriverManager –
Manages the list of database drivers

• Statement –
Used to submit SQL queries

• Connection –
Interface with all methods for creating a database.

• ResultSet –
Holds data retrieved from a database after executing a query.